

CAPSTONE PROJECT

Criminal Code violations 2020 – 2024 by Province and Territory

1. Data Processing

Shape before cleaning: (328, 167)

Shape after cleaning: (145, 160)

Columns selected for analysis:

['Total violent Criminal Code violations', 'Total property crime violations', 'Total other Criminal Code violations', 'Total Criminal Code traffic violations', 'Total Drug violations', 'Total Cannabis Act', 'Youth Criminal Justice Act', 'Total other Federal Statutes']

Offences dataset:

Dataset:

2. Machine Learning

Absolute Crime Levels - PCA Variance Explained:

PC1: 0.818

PC2: 0.066

Total (2 PCs): 0.884

Absolute Crime Levels - Cluster Size (Descriptive Labels):

cluster_label

High Crime 1

Low Crime 143

Name: count, dtype: int64

Absolute Crime Levels - Cluster Feature Means:

Absolute Levels - High Crime Locations (count=1):

Location ClusterLabel

70 Halifax Metropolitan Area, NS High Crime

Proportional Structure - PCA Variance Explained:

PC1: 0.165

PC2: 0.110

Total (2 PCs): 0.275

Proportional Structure - Cluster Size (Descriptive Labels):

```
cluster_label
Pattern 1   59
Pattern 2   84
Pattern 3    1
Name: count, dtype: int64
```

Proportional Structure - Cluster Feature Means:

Proportional Structure - High Crime Locations (count=84):

```
Location ClusterLabel
3   Sheshatshiu, NL  High Crime
4   Bay St. George, NL  High Crime
7   Burgeo, NL  High Crime
9   Roddickton, NL  High Crime
18  Bell Island, NL  High Crime
..   ...
139 St. Andrews, NB  High Crime
140 Bouctouche, NB  High Crime
141 Campbellton, NB  High Crime
142 Oromocto, NB  High Crime
143 Hampton, NB  High Crime
```

```
[84 rows x 2 columns]
```

Crime Growth Patterns - PCA Variance Explained:

```
PC1: 0.236
PC2: 0.189
Total (2 PCs): 0.425
```

Crime Growth Patterns - Cluster Size (Descriptive Labels):

```
cluster_label
Growth Group 1   143
Growth Group 2    1
Name: count, dtype: int64
```

Crime Growth Patterns - Cluster Feature Means:

```
      0      1      2      3      4      5 \
cluster_label
Growth Group 1 -0.033296 -0.307667 -0.054945 -0.050916 -0.019444 -0.074966
Growth Group 2  0.000000  0.000000  0.000000  0.000000  0.500000  0.666667

      6      7      8      9     10     11 \
cluster_label
Growth Group 1  0.148379  0.199297  0.014857  0.04136  0.233346  0.266525
Growth Group 2 -0.600000  1.000000 -1.000000  0.000000  0.000000  0.000000

     12     13     14     15     16     17 \
```

cluster_label							
Growth Group 1	0.053625	0.149298	0.112181	0.165011	0.507953	0.235328	
Growth Group 2	0.000000	0.000000	-0.500000	5.000000	0.000000	-1.000000	

	18	19	20	21	22	23 \	
cluster_label							
Growth Group 1	0.08859	0.090821	0.114198	0.079916	0.131298	0.012357	
Growth Group 2	0.00000	0.000000	14.000000	-0.400000	-0.222222	0.285714	

	24	25	26	27	28	29 \	
cluster_label							
Growth Group 1	0.123146	0.045668	0.114201	0.056053	-0.153613	-0.032051	
Growth Group 2	-0.666667	1.500000	0.000000	0.400000	0.000000	0.000000	

	30	31 cluster	
cluster_label			
Growth Group 1	-0.069294	-0.010185	0.0
Growth Group 2	0.000000	0.000000	1.0

Growth Patterns - High Crime Locations (count=1):

Location	ClusterLabel
126 Campobello, NB	High Crime

Silhouette Scores Summary:

Absolute Levels: 0.927 (n_clusters=2)

Proportional Structure: 0.124 (n_clusters=3)

Growth Patterns: 0.813 (n_clusters=2)